

FORGE — Full-Stack Product Specification

Fabrication & Operations Reconstruction & Guidance Engine

Version 1.0 | Authored for AI Agent Implementation

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1. EXECUTIVE VISION

1.1 Product Tagline

"Think it. Build it. Ship it." — The command center for mechanical fabrication engineers.

1.2 Core Problem Being Solved

Mechanical fabrication companies today receive engineering drawings (PDFs, CAD exports, Excel BOMs) from clients and spend days manually:

- Extracting dimensions, part numbers, materials, weld specs from PDFs
- Creating quotations in spreadsheets with manual BOM lookups
- Coordinating manufacturing plans across team members via email/paper
- Tracking revisions and sign-offs with no version history
- Producing output documents by copy-pasting across Word files

FORGE eliminates all of this. It is the Iron Man suit for mechanical engineers — an intelligent workspace that reads engineering documents, understands context, extracts all structured data, generates quotations, tracks manufacturing progress, and produces presentation-grade deliverables.

1.3 Target Users

- **Primary:** Mechanical Engineers, Fabrication Coordinators, Manufacturing Planners
- **Secondary:** Project Managers, QA Reviewers, Procurement Officers, Shop Floor Supervisors
- **Tertiary:** Company Admins, Clients (read-only portal, future scope)

1.4 Key Differentiators

- Context-aware PDF intelligence (not OCR copy-paste — true semantic extraction)
- Bring-Your-Own-AI model support (10+ providers)
- Tony Stark-style Anime.js exploded-view visualization of mechanical assemblies
- Full manufacturing lifecycle from document intake to shop floor scheduling
- Productizable: multi-tenant, subscription-based, white-labelable

2. DESIGN SYSTEM & UX PHILOSOPHY

2.1 Visual Identity

Theme: Futuristic dark industrial — think JARVIS HUD meets precision engineering.

Color Palette (Dark Theme — Default):

Background Primary:	#0A0E1A	(near-black navy)
Background Secondary:	#0F1629	(dark panel)
Background Card:	#141B2D	(elevated card)
Surface:	#1A2440	(interactive surface)
Border:	#1E2D4A	(subtle border)
Accent Primary:	#00D4FF	(electric cyan — primary CTA)
Accent Secondary:	#FF6B35	(burnt orange — warnings/alerts)
Accent Tertiary:	#7B2FBE	(deep violet — badges/tags)
Success:	#00E5A0	(mint green)
Warning:	#FFB800	(amber)
Error:	#FF3B5C	(crimson)
Text Primary:	#E8F0FF	(near-white)
Text Secondary:	#8B9CC4	(muted blue-grey)
Text Tertiary:	#4A5980	(dim)
Highlight:	#00D4FF22	(translucent cyan glow)

Color Palette (Light Theme):

Background Primary: #F0F4FF
Background Secondary: #E8EDF8
Background Card: #FFFFFF
Surface: #F5F8FF
Border: #D0D9EE
Accent Primary: #0066CC
Accent Secondary: #E84E1B
Text Primary: #0A1029
Text Secondary: #4A5568

2.2 Typography

Primary Font: "Inter" (UI, body text)
Monospace Font: "JetBrains Mono" (dimensions, codes, measurements)
Display Font: "Space Grotesk" (dashboard headers, KPI numbers)

2.3 Component Design Language

Cards: Subtle glassmorphism with 1px cyan border glow on hover. `backdrop-filter: blur(12px)`. Box shadow: `0 4px 24px rgba(0,212,255,0.08)`.

Buttons:

- Primary: Solid cyan with black text, slight glow on hover
- Secondary: Ghost with cyan border
- Destructive: Red border ghost
- AI Action: Violet gradient with pulse animation

Icons: Lucide React icons throughout. Custom SVG icons for manufacturing-specific concepts.

Inputs: Dark surface with cyan focus ring, animated label float.

Loading States: Scanning animation using CSS grid lines — like a radar sweep or HUD scan.

Transitions: Framer Motion for route transitions (slide + fade). Anime.js for data visualizations.

2.4 Layout System

Sidebar (collapsed: 64px / expanded: 260px) — persisted in user preferences
Top Bar (56px) — breadcrumb, user avatar, global search, notifications
Main Content Area — responsive, scrollable
Right Panel (optional, 320px) — context-aware AI assistant / detail pane

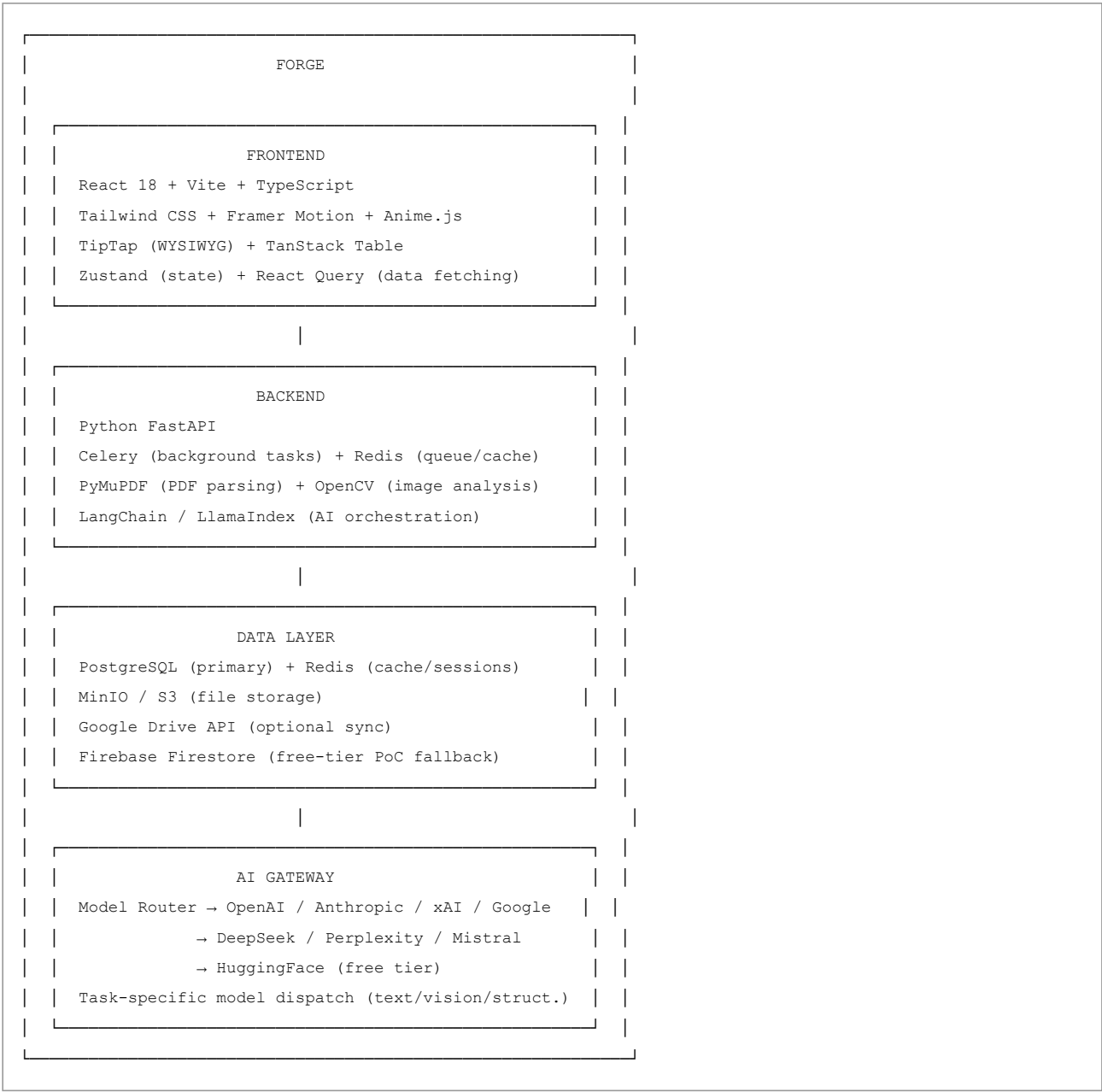
Grid: 12-column CSS Grid. Cards use 4/6/12 column spans based on viewport.

2.5 UX Principles

1. **Zero cognitive overhead on repeated tasks** — everything is remembered, autocompleted, suggested
2. **No dead ends** — every screen has a clear next action
3. **Progressive disclosure** — show minimal info first, expand on demand
4. **AI is ambient** — always available but never intrusive (right panel toggle)
5. **The dashboard is a command center** — KPIs visible in 3 seconds
6. **Every destructive action requires confirmation with consequence explanation**

3. SYSTEM ARCHITECTURE OVERVIEW

3.1 High-Level Stack



3.2 Deployment Model

- **Local/Self-hosted:** Docker Compose (all services)
- **Cloud:** Any VPS, Railway, Render, or AWS/GCP/Azure
- **Multi-tenant:** Each organization is isolated via `org_id` scoping on all DB queries
- **Environment config:** `.env` file for app-level keys; per-org keys stored encrypted in DB

4. AUTHENTICATION & ACCESS CONTROL

4.1 Authentication Flows

Option A: Google OAuth (if configured)

- Sign in with Google button on login page
- Google profile photo used as avatar
- Email auto-verified

Option B: Email + Password

- Standard email/password with bcrypt hashing
- Email verification on signup
- Forgot password via email link

Option C: 6-Digit Session Lock PIN

- After login, user can optionally set a 6-digit PIN
- PIN stored in browser localStorage (hashed with SHA-256)
- When screen is locked (idle timeout configurable: 5/10/15/30 min), PIN screen appears
- Reset PIN via security question (set during first PIN creation)
- Security question + answer stored in localStorage (answer hashed)
- PIN does NOT replace login — it is only a "screen lock" for when you step away

Implementation Notes:

- JWT access tokens (15 min expiry) + refresh tokens (30 day expiry) stored in httpOnly cookies
- CSRF protection on all state-changing endpoints
- Rate limiting on auth endpoints (5 attempts → 15 min lockout)

4.2 Role Hierarchy (Organization-wide)

Super Admin (app owner) > Org Admin > Owner > Coordinator > Contributor > Reviewer > Reader

Org Admin: Can create/delete projects, manage all members, configure org settings, billing **Owner (Project):** Full CRUD on specific project, can assign roles within project **Coordinator:** Can assign tasks, update statuses, view all workbooks **Contributor:** Can edit workbooks/sections they are assigned to **Reviewer:** Can add review comments, approve/reject sections, cannot edit content **Reader:** View-only access to projects they are invited to

4.3 Permission Matrix

Action	Reader	Reviewer	Contributor	Coordinator	Owner	Org Admin
View project	✓	✓	✓	✓	✓	✓
Edit workbook	—	—	✓	✓	✓	✓
Add review comments	—	✓	✓	✓	✓	✓
Approve sections	—	✓	—	✓	✓	✓
Create quotation	—	—	✓	✓	✓	✓
Publish revision	—	—	—	✓	✓	✓
Manage project members	—	—	—	—	✓	✓
Add to org inventory	—	—	—	✓	✓	✓
Configure org settings	—	—	—	—	—	✓
Manage subscriptions	—	—	—	—	—	✓

5. ORGANIZATION & TEAM MANAGEMENT

5.1 Organization Setup (First-Time Onboarding)

Step 1 — Welcome Screen

- "Create your organization" form: Org name, industry (manufacturing/fabrication/engineering), logo upload, country, timezone
- Option to join existing org via invite link

Step 2 — Initial Configuration

- Default currency, units (Imperial/Metric), fiscal year start
- Labour rate defaults (\$/hr — per trade: welder, fitter, painter, QA)
- Company address + branding colors (for output documents)

Step 3 — Invite Team

- Bulk invite by email with role assignment
- Invitation link (sharable, 7-day expiry, role pre-set)

After Setup: Organization screen is hidden. It lives only in **Settings > Organization**.

5.2 Organization Settings Page

- Org profile: name, logo, address, contact
- Branding: primary/secondary color, font for output documents
- Default units, currency, timezone
- Labour rate table (per trade, configurable)
- Overhead/markup defaults
- Team members management (add/remove/change roles)
- Invitation management (active invites, resend, revoke)
- Subscription & billing
- API key management (for app-level keys used by org)
- Storage configuration (local, Google Drive, S3)

5.3 Member Invitation Flow

- Inviter selects role + optional project access
- Email sent with link + temp access code
- Invitee creates account (or links existing) → auto-joined to org with assigned role
- New member onboarding checklist shown on first login

6. AI CONFIGURATION & MODEL REGISTRY

6.1 AI Gateway Architecture

The AI Gateway is a central router that dispatches tasks to the appropriate model based on:

1. Task type (see 6.2)
2. User/org configuration
3. Fallback chain if primary fails

6.2 Task-Type to Model Mapping

TASK TYPE	RECOMMENDED	FALLBACK
pdf_parse_structure	Claude 3.5 Sonnet	GPT-4o, Gemini 1.5
pdf_extract_dimensions	GPT-4o Vision	Claude 3.5, Gemini
text_generation_standard	GPT-4o-mini	DeepSeek, Mistral
text_generation_premium	GPT-4o / Claude Opus	Gemini 1.5 Pro
bom_extraction	Claude 3.5 Sonnet	GPT-4o
cad_instruction_gen	GPT-4o	Claude 3.5
quotation_calculation	GPT-4o-mini	DeepSeek
search_and_citations	Perplexity Sonar	Tavily + GPT-4o
creative_document_format	Claude 3.5 Sonnet	GPT-4o
3d_model_gen (future)	Point-E / Shap-E	Custom
free_tier_fallback	HuggingFace Mistral	HuggingFace Zephyr

6.3 Supported AI Providers

Provider	Models	Key Storage
OpenAI	GPT-4o, GPT-4o-mini, o1	.env + org DB
Anthropic	Claude 3.5 Sonnet, Claude Opus	.env + org DB
xAI	Grok-2	.env + org DB
Google	Gemini 1.5 Pro/Flash	.env + org DB
Perplexity	Sonar Pro/Online	.env + org DB
DeepSeek	DeepSeek-V3	.env + org DB
Microsoft	Azure OpenAI	.env + org DB
Mistral	Mistral Large/Small	.env + org DB
HuggingFace	Mistral-7B, Zephyr, Phi	Free (Hub API)
Ollama	Any local model	localhost config

6.4 Key Management Rules

- 1. App-level keys set in .env → used for orgs on admin-assigned plans (demo/trial)
- 2. Org-level keys set in Settings → override app keys for that org
- 3. User-level keys (personal) → override for personal experiments (not used in shared workbooks)
- 4. Keys are AES-256 encrypted before DB storage; decrypted only at request time in memory
- 5. Keys are never exposed in frontend responses

6.5 AI Settings UI

Location: Settings > AI Configuration

- **Global model selector:** Per task-type dropdown (see 6.2 table)
- **Provider cards:** Show connection status (green/red), remaining quota if available via API
- **Test connection button:** Sends a test ping to verify key works
- **Free tier section:** HuggingFace Hub login OR anonymous (rate limited)
- **"Use same key for all tasks" toggle:** Simplifies setup for single-provider users
- **Cost tracker:** Estimated token spend this month (requires provider API)

7. STORAGE & INTEGRATION LAYER

7.1 File Storage Options (Priority Order)

Tier 1 — Configured Cloud Storage

- AWS S3 / compatible (MinIO for self-hosted)
- Google Cloud Storage

Tier 2 — Google Drive (if configured)

- OAuth 2.0 Google Drive API
- Folder structure: `FORGE/{org_name}/{project_name}/{workbook_name}/`
- Files synced automatically on save
- Read from Drive: browse Drive folders in file picker

Tier 3 — Free-tier PoC Storage

- **Firebase Firestore:** Document metadata + small payloads
- **Google Sheets via API:** Structured data (BOM, inventory lists, quotations)
- Files (PDFs, images): Firebase Storage (free tier: 5GB)

Tier 4 — Local/Browser

- IndexedDB for unsaved drafts
- File cache for recent uploads

7.2 Storage Configuration UI

Location: Settings > Storage

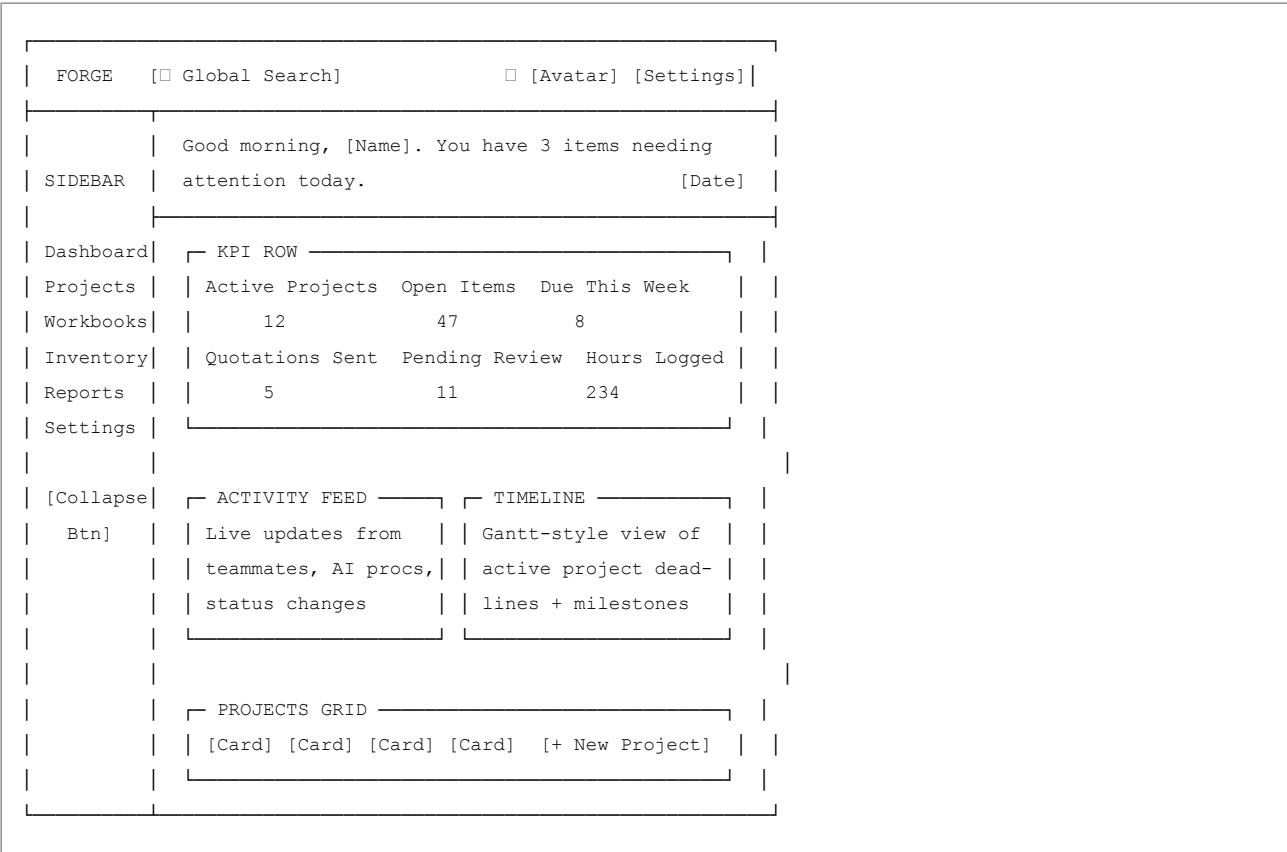
- Storage provider selector (radio buttons with status indicators)
- Connection credentials form per provider
- Test connection + quota display
- Google Drive OAuth button ("Connect Drive")
- Auto-sync toggle: On save → sync to configured storage
- Backup schedule (daily/weekly — creates ZIP archive)

7.3 Data Persistence Formats

- **Primary:** PostgreSQL + file storage for production
- **PoC/Dev:** JSON files in `/data/` directory OR Firebase
- **Serialization:** All domain objects have `to_json()` / `from_json()` methods
- **Pickle/Shelve:** Available for offline/dev mode session persistence

8. DASHBOARD

8.1 Dashboard Layout (Command Center)



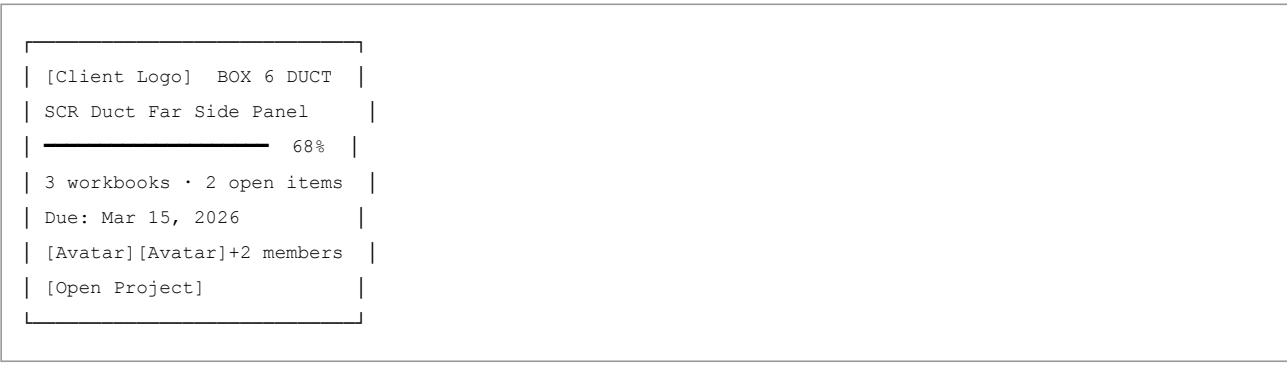
8.2 KPI Cards

Each KPI card: animated number counter on load (Anime.js), color-coded status ring, click to drill down.

8.3 Activity Feed

- Real-time (WebSocket or polling every 30s)
- Events: workbook status change, review comment added, revision published, section approved, teammate joined
- Each event has icon, actor avatar, timestamp, link to item
- Filter by: All / My Items / Projects I Own / Mentions

8.4 Project Cards (Grid View)



9. PROJECT MANAGEMENT MODULE

9.1 Creating a New Project

Project Creation Modal/Page:

Project Name*	[Text input]
Project Description	[Rich text – brief]
Client Name	[Text – autocomplete from org history]
Client PO Number	[Text]
Internal Reference #	[Text – auto-generated or manual]
Project Type	[Dropdown: Fabrication / Assembly / Design / Mixed]
Start Date*	[Date picker]
Target Completion Date*	[Date picker]
Priority	[Low / Medium / High / Critical]
Tags	[Multi-select + create new]
Assign Team Members	[Multi-select users from org + set roles]
Project Inventory	[Link/create project-specific inventory]
Base Currency	[Inherit from org or override]
Unit System	[Inherit from org or override]
Cover Image	[Upload or auto-generate from project type]

9.2 Project Dashboard

Sections:

1. **Header:** Project name, client, status badge, % complete ring, key dates
2. **Summary Cards:** Total workbooks, sections, open reviews, overdue items
3. **Workbooks List:** Sortable table with status, assignee, last modified, completion %
4. **Gantt/Timeline:** Drag-and-drop timeline for workbook milestones
5. **Financial Summary:** Total estimated cost, approved quotations, variance
6. **Team Panel:** Member avatars, roles, recent activity
7. **Notes/Log:** Project-level notes (WYSIWYG, timestamped entries)

9.3 Project-Level Inventory

- Separate inventory scope from org-level
- Project parts pre-loaded from BOM (if provided at project creation)
- Editable: add/remove/update parts, pricing, suppliers
- Downstream: referenced in all workbooks within the project
- Export: CSV, Excel

9.4 Project Status Flow

Planning → Active → In Review → Complete → Archived

Each transition requires confirmation. Archived projects are read-only.

10. WORKBOOK ENGINE

10.1 Workbook Concept

A **Workbook** represents one engineering drawing file and all the work done around it. Each project can have multiple workbooks (e.g., one per panel, assembly, or drawing revision).

10.2 Creating a New Workbook

Workbook Creation Flow:

Step 1: Basic Info

- Workbook Name*
- Description
- Drawing Number (from PDF – auto-detected or manual)
- Revision (from PDF – auto-detected)
- Tags
- Assigned Members (inherit from project or override)
- Timeline: Start date, Target completion

Step 2: File Upload

- Drag & drop zone for PDF (primary)
- Additional context files: Excel BOM, CSV specs, images, other PDFs
- "Analyze Document" button → triggers PDF Intelligence Pipeline (Section 11)

Step 3: Initial Analysis (auto + review)

- System shows: detected sections, page count, extracted metadata
- User reviews, confirms, or corrects the section breakdown
- "Initialize Workbook" → creates sections

10.3 Workbook Dashboard

WORKBOOK HEADER

[Drawing thumbnail]	CPF#8250016-6B1 Rev.2
SCR Duct Far Side Panel – Box 6	
Status: [PLAN STARTED ▼]	Revision: R3 Last saved: 2 min
Assigned: [Avatars]	Due: Mar 10 Progress: 52%

TABS: [Overview] [Sections] [BOM] [Specs] [Schedule] [Quotation] [Outputs] [Revisions]

10.4 Workbook Status Flow

Draft → Quotation Complete → Approved → Plan Started →
Plan Complete → Design Complete → Review Complete → Done

Visual indicator: Horizontal stepper at top of workbook. Each step is clickable (with confirmation if going backwards). Color: grey → yellow → cyan → green at Done.

10.5 Section Tabs Within Workbook

Overview Tab:

- Summary card (AI-generated, user-editable)
- Key specs extracted from drawing (table view)
- Timeline widget
- Pending actions checklist

Sections Tab:

- List of all sections detected from PDF
- Each section card: thumbnail of PDF cutout, status, spec count, last updated

- Status per section: Draft / In Progress / Review / Complete

BOM Tab:

- Full Bill of Materials table (editable, from PDF extraction + manual additions)
- Columns: Item #, Part Name, Description, Material, Qty, Unit, Unit Cost, Total, Supplier, Notes
- Total cost rollup
- Export to Excel / PDF

Specs Tab:

- All extracted specifications, codes, notes, weld symbols
- Organized by: General Notes / Section-Specific / Standards Referenced
- Each spec has source reference (section + page)

Schedule Tab:

- Task list for manufacturing (see Section 10.7)
- Assignees, due dates, dependencies
- Progress tracking

Quotation Tab: (see Section 13)

Outputs Tab: (see Section 18)

Revisions Tab: (see Section 16)

10.6 Section Cards

Each Section Card (expanded view):

SECTION: SECTION A-A (TYP. 9 PLCS)	
[PDF Cutout – zoomable image]	Status: [IN PROGRESS ▼]
	Assigned: [Avatar] John D.
	Last Rev: R2 → R3 (Feb 17)
AI SUMMARY	
"Cross-sectional weld assembly at 9 typical locations. CSG plate welded at 3/16" fillet, stiffeners tacked only. Pipe reference: 6-CSG9, 6-CSG8, 6-CSG7. 1" CSG gap maintained."	
EXTRACTED SPECS	
Weld size	3/16" fillet (typical)
CSG gap	1"
Tack only zones	All stiffeners (no full weld)
Standard	W59 / AWS D1.1 / AWS D1.6
Typical qty	9 places
REVIEW COMMENTS (2) NOTES (1) AI SUGGESTIONS CITATIONS	
[Export Parallax View] [Add Note] [Request Review] [Mark Done]	

10.7 Manufacturing Schedule (per Workbook)

Tasks generated by AI from workbook sections, editable by coordinators:

Task	Trade	Est. Hours	Assigned	Start	End	Dependencies	Status
Material Procurement	PM	8h	Alice	Mar 1	Mar 3	—	Planned
Plate Cutting	Operator	16h	Bob	Mar 4	Mar 5	Procurement	Planned
Fit-up (Box Stage)	Fitter	24h	Charlie	Mar 6	Mar 8	Cutting	Planned
Welding — CSG	Welder	32h	Dave	Mar 9	Mar 13	Fit-up	Planned
Weld QA Inspection	QA	8h	Eve	Mar 14	Mar 14	Welding	Planned
Painting	Painter	12h	Frank	Mar 15	Mar 16	QA	Planned
Final Inspection	QA	4h	Eve	Mar 17	Mar 17	Painting	Planned
Shipping	Logistics	4h	Alice	Mar 18	Mar 18	Inspection	Planned

11. PDF INTELLIGENCE PARSER

11.1 Philosophy

Not OCR copy-paste. The parser uses vision AI models to semantically understand mechanical drawings — extracting structured data (dimensions, parts, weld symbols, notes) by understanding *context*, not just reading text.

11.2 Parsing Pipeline

PDF File Input

|



Stage 1: Pre-processing

- └ Extract all pages as high-res images (PyMuPDF @ 300 DPI)
- └ Extract embedded text with coordinates (PyMuPDF)
- └ Extract bookmarks/outline tree
- └ Detect page orientation + scale
- └ Store raw page data + coordinates

|



Stage 2: Page Structure Analysis (Vision AI)

- └ Identify drawing areas vs. title blocks vs. notes sections
- └ Detect section dividers (callout boxes, labels like "SECTION A-A")
- └ Map text to regions (dimensions belong to which drawing area?)
- └ Extract BOM/parts list tables if present
- └ Return: structured page layout JSON

|



Stage 3: Section Extraction

- └ Crop each identified section from page image
- └ Extract: section name, scale, view type (elevation/section/detail)
- └ For each section: extract all text with spatial context
- └ Output: {section_id, page, bounds, cropped_image_url, raw_text}

|



Stage 4: Semantic Extraction (per section)

- └ DIMENSIONS: all measurements with units and associated parts
- └ PART REFERENCES: part numbers, quantities, material codes
- └ WELD SYMBOLS: decode weld specification symbols (AWS A2.4)
- └ NOTES: general notes, special instructions, material callouts
- └ STANDARDS: referenced codes (AWS D1.1, W59, ASME, ISO etc.)
- └ TOLERANCES: dimensional tolerances, GD&T callouts
- └ REVISION NOTES: changes documented in revision block

|



Stage 5: Cross-Reference & Validation

- └ Match part numbers to project/org inventory
- └ Flag inconsistencies (same part with different dims)
- └ Verify BOM quantities match drawing callouts
- └ Generate confidence scores per extracted data point

|



Stage 6: Section Assembly

- └ Generate AI summary per section
- └ Create initial spec table per section
- └ Suggest manufacturing tasks based on content
- └ Output: complete workbook structure ready for user review

11.3 Model Dispatch for Parsing

```
PARSER_MODEL_CHAIN = {
    "page_structure": ["claude-3-5-sonnet", "gpt-4o", "gemini-1.5-pro"],
    "dimension_extract": ["gpt-4o-vision", "claude-3-5-sonnet"],
    "bom_table_extract": ["claude-3-5-sonnet", "gpt-4o"],
    "weld_symbol_decode": ["gpt-4o-vision", "claude-3-5-sonnet"],
    "summary_generation": ["claude-3-5-sonnet", "gpt-4o-mini", "mistral-7b"],
    "fallback_ocr": ["tesseract", "easyocr"] # if AI unavailable
}
```

11.4 Weld Symbol Decoder

Mechanical drawings contain AWS weld symbols that encode multiple specifications in a single symbol. FORGE decodes these automatically:

- Arrow side / other side
- Weld type (fillet, groove, plug, seam, spot)
- Size, length, pitch
- Contour (flush, convex, concave)
- Finish symbol (G=grind, M=machine, C=chip)
- Field weld indicator
- All-around weld indicator

Output: human-readable weld specification string + structured JSON

11.5 Drawing Standards Recognition

FORGE recognizes and contextualizes references to:

- **AWS:** D1.1, D1.6, A2.4 (Welding)
- **ASME:** B31.1, B31.3, B16.5 (Pressure piping)
- **AISC:** 360 (Steel construction)
- **ASTM:** Material standards (A36, A572, etc.)
- **ISO:** 9013, 9692 (Cutting/weld prep)
- **CSA/W:** W59 (Canadian welding standard — detected in sample PDF)
- **NDE/NDT:** RT, UT, MT, PT methods

12. SECTION ANALYSIS & EXTRACTION ENGINE

12.1 Section Work Interface

The section work interface is the "cockpit" where engineers work through each extracted section.

Layout (3-panel):

LEFT: PDF VIEWER	CENTER: WORK AREA	RIGHT: AI PANEL
[Zoomable PDF cutout of this section]	Section Summary	AI Assistant
[Zoom +/-]	Spec Table	Suggestions
[Fit/Full]	Notes	Citations
[Pan]	Review Comments	Standards Lookup
[Annotate]	Related Parts (BOM)	Similar Sections
Section: A-A / D-D / B-B...	Actions	[Ask AI...]
[< Prev] [Next >]		

12.2 PDF Viewer Features

- Pan and zoom (mouse wheel + trackpad + buttons)
- Annotation overlay: draw boxes, add sticky notes, highlight dimensions
- "Ghost overlay" mode: show extracted data overlaid on the PDF with colored boxes
- Measurement tool: click two points to verify a dimension
- Side-by-side: compare with previous revision

12.3 Spec Table

- Editable table: all extracted specifications for this section
- Columns: Parameter, Extracted Value, Verified Value, Unit, Notes, Source Reference, Confidence
- Confidence badge: High (green) / Medium (yellow) / Low (red — requires human verification)
- Click any row → highlights the corresponding area in PDF viewer
- Add row manually; delete with confirmation
- AI re-check button per row

12.4 Review Comments System

- Threaded comments per section
- Comment types: Question / Issue / Suggestion / Approval
- Status: Open / In Progress / Resolved
- @mention team members → notification sent
- Link to specific spec row or PDF annotation
- File attachments in comments
- Reviewer can mark section as "Approved" or "Changes Required"

12.5 AI Feedback & Suggestions

- **Flagged anomalies:** "Weld size X seems undersized for material thickness Y — verify against AWS D1.1 Table 5.8"
- **Missing specs:** "No surface finish specified for this joint — typically required for this material grade"
- **Code citations:** Auto-fetched from Perplexity/web if configured, else from internal standards database
- **Similar section lookup:** "This matches Section D-D in Workbook SCR-001 — compare?"
- **AI rewrite:** Selected text in notes/summary → "Rewrite more clearly / more technical / shorter"

13. QUOTATION & COST ESTIMATION MODULE

13.1 Quotation Types

Type A: Detailed Study Quotation

- Built from extracted BOM + section specs
- Each line item: material, labour, process with cost breakdown
- Best for: when full drawing analysis is complete

Type B: Blank/Preliminary Quotation

- Based only on provided client BOM (if any)
- Estimates from org-level labour rates + material price database
- Flags: "Subject to detailed review — contingency applied"
- Best for: quick turnaround before full analysis

13.2 Cost Components

1. Material Costs

- From BOM: Part description, material grade, quantity, weight (kg/lb), unit cost (\$/kg or \$/unit)
- Steel price database (configurable, auto-updated if API available): A36, A572, SS304, SS316, etc.
- Weight calculated from dimensions if available (density × volume)
- Scrap/wastage factor: configurable (default 10%)

2. Labour Costs

- Task list (from manufacturing schedule — Section 10.7)
- Per task: trade, estimated hours, hourly rate → subtotal
- Labour rate source: org defaults or project override
- Productivity factor: configurable (accounts for job complexity)
- Overtime allowance: toggle + multiplier

3. Sub-operations

- Laser/plasma cutting: length (m) × cost/m
- Welding consumables: wire/rod quantity + cost
- Heat treatment: type + unit cost
- Surface preparation: area (m²) × cost/m²
- NDT/Inspection: method + hours + rate
- Machining: hours × rate
- Hardware/fasteners: from BOM

4. Painting/Coating

- Surface area calculated from drawings
- Paint system specification (primer, intermediate, topcoat)
- DFT requirements
- Area × cost/m² per coat

5. Testing & Inspection

- Hydrostatic test (if applicable)
- Dimensional inspection report
- Material traceability documentation
- Third-party inspection fees

6. Overheads & Markup

- Fixed overhead: % of direct costs (configurable at org/project level)
- Profit margin: % (configurable, separate for labour vs material)

- Contingency: % buffer (recommended 10-15% for fabrication)
- Freight/shipping estimate
- Insurance/bonds (if required)

13.3 Quotation Builder UI

QUOTATION: QT-2026-0047

Project: CPF#8250016-6B1 | Prepared by: John D. | Date: Feb 21, 2026

SUMMARY

Material:	\$45,230	52%
Labour:	\$28,400	33%
Sub-operations:	\$4,800	5.5%
Painting:	\$3,200	3.7%
Testing:	\$1,500	1.7%
Overhead (12%):	\$9,998	
Contingency(10%):	\$9,313	
TOTAL ESTIMATE:	\$102,441	

[Detailed breakdown tabs] [Edit any line] [AI Validate] [Export]

13.4 Quotation Document Output

- Company letterhead (from org branding)
- Professional formatting with tables and section breakdowns
- Terms & conditions (template, editable per org)
- Validity period, payment terms
- Exclusions list
- Export: PDF (primary), DOCX, Excel
- Direct email to client (if email configured)

14. INVENTORY MANAGEMENT SYSTEM

14.1 Inventory Levels

Organization-level Inventory: Master catalog for all org projects **Project-level Inventory:** Project-specific items, prices, suppliers

Items at project level override org-level for that project. Items can be "promoted" from project → org level with permissions.

14.2 Inventory Item Schema

```
{
  "id": "uuid",
  "org_id": "uuid",
  "project_id": "uuid | null",
  "part_number": "6-CSG8",
  "description": "1/4\" Plate 44W",
  "category": "Plate",
  "material_grade": "ASTM A36 / 44W",
  "dimensions": {
    "length_mm": 10338,
    "width_mm": 2921,
    "thickness_mm": 6.35
  },
  "weight_kg": 3290,
  "unit": "EACH",
  "unit_cost": 4.20,
  "currency": "USD",
  "supplier": "Steel Co.",
  "supplier_part_no": "PL-44W-1/4",
  "lead_time_days": 5,
  "min_order_qty": 1,
  "current_stock": 0,
  "notes": "Grade 44W per CSA G40.21",
  "tags": ["plate", "structural", "44w"],
  "created_by": "uuid",
  "updated_at": "ISO8601"
}
```

14.3 Inventory Features

- **Search & filter:** Full-text search, filter by category/material/supplier
- **Autocomplete:** Used in BOM editor, quotation builder — type part number → autocomplete from inventory
- **Price history:** Track price changes over time with dates
- **Supplier management:** Multiple suppliers per item with preferred supplier flag
- **Import:** CSV/Excel import with column mapping wizard
- **Export:** BOM template, procurement list
- **Low stock alerts:** Set minimum stock levels → notification when below
- **Usage tracking:** Shows which projects/workbooks use this item

15. WYSIWYG EDITOR & SMART TABLES

15.1 WYSIWYG Editor (TipTap)

Used for: Section summaries, quotation narratives, project notes, output documents.

Features:

- Bold, italic, underline, strikethrough
- Headings H1-H4
- Bullet and numbered lists
- Code blocks (for part codes, standards)
- Blockquote (for cited specs)
- Tables (basic — see Smart Tables below for advanced)

- Image insertion
- Link insertion with preview
- Divider
- Text color and highlight
- **AI Toolbar (inline):** Select text → AI options appear:
 - Rewrite (general)
 - Make more technical
 - Make more concise
 - Fix grammar
 - Translate (future)
 - Suggest citation

15.2 Smart Tables (TanStack Table)

For: BOM, Spec tables, Quotation line items, Inventory, Schedule

Features:

- Inline cell editing (click to edit)
- Column: sort, filter, resize, hide, reorder (drag)
- Row: add, delete, duplicate, reorder
- Multi-row select + bulk operations
- Column formulas: e.g., $\text{Weight} \times \text{Price} = \text{Total Cost}$
- Cell types: text, number, currency, date, dropdown, autocomplete (from inventory), checkbox, file link
- Row grouping (group by material, section, etc.)
- Freeze header row + first column
- Search/filter row above table
- Export: CSV, Excel, PDF
- Pagination or infinite scroll
- Conditional formatting: e.g., high-cost items highlighted red
- Copy/paste from Excel (tab-separated paste)
- Undo/redo (Ctrl+Z)

16. REVISION CONTROL SYSTEM

16.1 Revision Concepts

A **Revision** is a saved snapshot of the entire workbook state, including: all section specs, BOM, notes, quotation, schedule. Think of it like Git commits for engineering documents.

16.2 Revision Flow

```
R0 (Initial) → R1 → R2 → R3 (Current Active)
                ↑
            (User can activate any past revision)
```

Creating a Revision:

- Click "Save as Revision" button
- Enter: Revision description, type (Minor/Major/Correction)
- System captures: timestamp, author, diff from previous revision
- Revision is immutable after creation (cannot overwrite)

Viewing Revisions:

- Timeline view: horizontal bar showing R0 → current
- Each revision: author avatar, date, description, change count
- Diff viewer: side-by-side comparison of spec tables, notes, BOM

Activating a Past Revision:

- User selects older revision → click "Activate as Working Revision"
- **Warning dialog:** "Activating R1 will create R4 as a branch from R1. Revisions R2, R3 will still be visible but become archived (read-only). Any new changes will create R4." Checkbox: "I understand"
- If user wants to retain R2/R3 as editable: "Lock R2/R3" button → marks them as archived reference, creates new branch

Revision Lock:

- Any revision can be marked "Locked (Read-only)" — no new edits, preserved for reference
- Only Owner/Admin can lock/unlock revisions

16.3 Change Tracking

- All spec table changes are logged with: field, old value, new value, user, timestamp
- PDF annotations are versioned
- Comments preserve their revision context

17. PARALLAX VISUALIZATION ENGINE (ANIME.JS)

17.1 Vision

Inspired by the anime.js website's stunning product exploded views — FORGE generates interactive, parallax-scroll-driven 3D decompositions of mechanical assemblies directly from the extracted section data.

Think: each component of an assembly floating in space, labeled with specs, revealed as you scroll, with the same cinematic quality as the anime.js homepage.

17.2 Visualization Levels

Section Level: Exploded view of one section's components **Workbook Level:** All sections in an assembly, progressively revealed **Project Level:** All workbooks as connected assemblies

17.3 AI-Generated Anime.js Visualization

Trigger: User clicks "Export Parallax View" on a section/workbook/project.

Process:

1. AI reads section data (component list, dimensions, relationships, weld connections)
2. AI generates an Anime.js JavaScript scene:
 - Each component as a styled `<div>` or SVG shape, sized proportionally
 - Components start as an assembled view, then "explode" outward on scroll
 - Each component labeled with part number, material, dimensions
 - Weld lines shown as animated dashed lines between components
 - Color-coded by material type
3. Scene is rendered in a fullscreen overlay or new tab
4. User can: pause, replay, zoom in on component, click component for spec popup

Output: Self-contained HTML file with embedded Anime.js — sharable, presentable without FORGE.

17.4 Visualization Spec Template (for AI generation)

When AI generates the Anime.js visualization, it receives a structured prompt like:

```
{
  "section_name": "SECTION A-A",
  "components": [
    {"id": "6-CSG9", "type": "plate", "dims": "407\" x 19\" x 1/4\"", "material": "44W", "color": "#4A90D9"},
    {"id": "6-CSG8", "type": "plate", "dims": "407\" x 115\" x 1/4\"", "material": "44W", "color": "#4A90D9"},
    {"id": "6-8", "type": "flatbar", "dims": "19-1/8\" x 7\" x 1/2\"", "material": "44W", "color": "#7ED321"}
  ],
  "connections": [
    {"from": "6-CSG8", "to": "6-CSG9", "type": "fillet_weld", "size": "3/16\""},
    {"from": "6-CSG8", "to": "6-8", "type": "tack_only"}
  ],
  "annotations": ["TYP. 9 PLCS", "1\" CSG GAP", "3/16 TYP."]
}
```

17.5 Parallax Scroll Behavior

Following the anime.js website design pattern:

- Scroll position drives animation progress (using IntersectionObserver + scroll event)
- Components fly in from off-screen edges at scroll threshold
- Assembly step labels appear with fade-in at each scroll position
- Spec callout lines extend with SVG path animation (stroke-dashoffset)
- "Fully assembled" view at the end with all specs visible
- Keyboard navigation: Arrow keys step through stages

18. OUTPUT & EXPORT MODULE

18.1 Output Types Per Workbook

1. Design Plan Document

- Full workbook as a formatted document
- Cover page: company logo, project name, drawing number, revision, date
- Table of contents (auto-generated)
- Section-by-section: PDF cutout + specs + notes + review summary
- BOM table
- Manufacturing schedule
- Revision history
- Custom branding (fonts, colors, header/footer — paid feature for advanced customization)
- Export: DOCX, PDF

2. Bill of Materials (BOM) Export

- Excel or CSV
- Columns: Item #, Part #, Description, Material, Qty, Unit, Unit Weight, Total Weight, Unit Cost, Total Cost, Supplier
- Subtotals by category
- FORGE watermark on free tier / company branding on paid

3. Quotation Document (see Section 13.4)

4. Purchasing Checklist

- Derived from BOM + quotation
- Format: Procurement-ready checklist
- Columns: Item, Spec, Supplier, PO #, Required Date, Status, Notes
- Export: PDF, Excel

5. Manufacturing Plan / Work Order

- For shop floor use
- Task-by-task breakdown with: operation, materials required, tools/equipment, inspection criteria, sign-off boxes
- Large-print format option (for shop floor display)

6. Spec Code Sheet

- All standards, codes, special instructions extracted from drawing
- Organized by: welding specs / material specs / finishing specs / testing requirements
- Ideal for QA/QC review

7. Parallax Presentation (see Section 17) — HTML export

18.2 Export Settings

- Company logo: pulled from org settings
- Font and color scheme: org branding
- Page format: A4 / Letter / A3
- Include/exclude sections (checkbox per section)
- "Draft" watermark toggle
- Password protection for PDF (paid)
- Digital signature field (paid, future)

19. NOTIFICATION & COLLABORATION SYSTEM

19.1 Notification Types

Event	Recipients
Workbook status changed	All project members
Review comment added	Assignee + @mentioned users
Review comment resolved	Comment author
Section approved	Workbook assignee
Revision published	All project members
Deadline approaching (7 days)	Assignee + Coordinator
Deadline overdue	Assignee + Owner
New member joined project	Project Owner
AI analysis complete	Initiating user
Quotation approved	Quotation author + Project Owner

19.2 Notification Delivery

- **In-app:** Bell icon with badge count; notification drawer
- **Email:** Configurable per notification type (opt-in per user)
- **Browser push:** Optional (requires user permission)

19.3 Collaboration Features

- **Presence indicators:** Small avatars on workbooks/sections showing who is currently viewing
- **"Typing" indicator:** In review comments thread
- **Live status updates:** Workbook status changes broadcast to all viewers in real-time (WebSocket)

- **@mention:** In any comment/note field — autocomplete from project members
 - **Shared cursors (future):** For live co-editing in sections
-

20. ADMIN PANEL & SUBSCRIPTION MANAGEMENT

20.1 App-Level Admin Panel

Access: Superadmin only (set in `.env`)

Sections:

- **Organizations:** List all orgs, usage stats, subscription tier, last active
- **Users:** Global user list, impersonation (for support), disable/enable
- **Subscriptions:** Create/modify plans, assign orgs to plans, set trial periods
- **AI Keys:** Set app-level AI keys for trial/demo orgs
- **Feature Flags:** Enable/disable features per org or globally
- **Analytics:** DAU, workbooks created, AI calls per day, storage used
- **System:** DB health, queue status, error logs, backup status

20.2 Subscription Plans (Recommended)

Free (Community)

- 1 organization, 3 users, 2 active projects
- 5 workbooks per project
- 10 AI calls/day (HuggingFace models only)
- 500MB storage
- FORGE branding on exports
- No custom branding

Starter (\$49/mo per org)

- 10 users, 10 active projects
- Unlimited workbooks
- 100 AI calls/day (bring your own key for premium models)
- 5GB storage
- Custom company logo on exports
- Google Drive integration

Professional (\$149/mo per org)

- 25 users, unlimited projects
- Unlimited AI calls (own keys)
- 25GB storage
- Full custom branding on exports
- Parallax visualizations
- Advanced quotation tools
- Priority support

Enterprise (Custom)

- Unlimited users & projects
- Dedicated infrastructure option
- White-label (custom domain, complete branding removal)

- SLA + dedicated support
- Custom AI model integration
- SSO (SAML)
- Audit log export
- On-premises deployment option

20.3 Feature Flags System

```
FEATURE_FLAGS = {  
    "parallax_visualization": ["professional", "enterprise"],  
    "advanced_quotation": ["starter", "professional", "enterprise"],  
    "google_drive_sync": ["starter", "professional", "enterprise"],  
    "custom_branding_export": ["professional", "enterprise"],  
    "white_label": ["enterprise"],  
    "ai_premium_models": ["starter", "professional", "enterprise"],  
    "bulk_export": ["professional", "enterprise"],  
    "revision_compare": ["starter", "professional", "enterprise"],  
    "client_portal": ["enterprise"],  
    "api_access": ["professional", "enterprise"]  
}
```

21. SETTINGS & PERSONALIZATION

21.1 User Profile Settings

- Display name, job title, department
- Profile photo (upload or from Google)
- Email address (change with verification)
- Password change (if email/password auth)
- Notification preferences (per type: in-app / email / push)
- Linked accounts: Google, GitHub (for Drive access)
- Session lock PIN setup (6-digit + security question)
- Idle lock timeout (5 / 10 / 15 / 30 min / Never)
- Language (future internationalization)

21.2 Appearance Settings

- Theme: Dark (default) / Light / System
- Sidebar: Auto-collapse / Always open / Always collapsed
- Density: Comfortable / Compact / Spacious
- Font size: Small / Medium / Large
- PDF viewer: Default zoom level

21.3 AI Preferences

- Preferred AI provider per task type
- API key configuration (per provider)
- AI assistant: Auto-suggest ON/OFF per section
- Citation style: IEEE / simple URL / disabled

21.4 Organization Settings (Org Admin only — see Section 5.2)

22. DATA MODELS & SCHEMA

22.1 Core Entities

```

-- Organizations
organizations (id, name, logo_url, address, currency, unit_system,
               fiscal_year_start, labour_rates_json, branding_json,
               subscription_tier, created_at)

-- Users
users (id, org_id, email, password_hash, display_name, avatar_url,
       role, google_id, preferences_json, created_at, last_active)

-- Projects
projects (id, org_id, name, description, client_name, client_po,
          internal_ref, project_type, status, start_date, target_date,
          priority, tags, created_by, created_at)

-- Project Members
project_members (project_id, user_id, role, added_by, added_at)

-- Workbooks
workbooks (id, project_id, name, description, drawing_number,
           revision_code, status, timeline_json, created_by, created_at)

-- PDF Uploads
pdf_uploads (id, workbook_id, filename, file_url, file_size,
             page_count, parsed_at, parse_status, parse_metadata_json)

-- Sections
sections (id, workbook_id, pdf_upload_id, name, section_type,
          page_number, bounds_json, cutout_image_url, status,
          assigned_to, ai_summary, created_at)

-- Spec Items
spec_items (id, section_id, parameter, extracted_value, verified_value,
            unit, notes, source_ref, confidence, row_order, created_by)

-- BOM Items
bom_items (id, workbook_id, section_id, item_number, part_number,
           description, material, quantity, unit, unit_weight,
           unit_cost, supplier, notes, inventory_item_id)

-- Comments
comments (id, section_id, parent_id, author_id, comment_type,
          content, status, spec_item_id, pdf_annotation_ref,
          created_at, resolved_at, resolved_by)

-- Revisions
revisions (id, workbook_id, revision_number, description, type,
           author_id, snapshot_json, diff_json, is_active,
           is_locked, created_at)

-- Quotations
quotations (id, workbook_id, type, status, line_items_json,
            totals_json, markup_json, validity_days, notes,
            created_by, approved_by, created_at)

```

```
-- Inventory Items
inventory_items (id, org_id, project_id, part_number, description,
                 category, material_grade, dimensions_json, weight_kg,
                 unit, unit_cost, currency, supplier, stock_qty,
                 lead_time_days, tags, created_by, updated_at)

-- Schedule Tasks
schedule_tasks (id, workbook_id, name, trade, estimated_hours,
                assigned_to, start_date, end_date, dependencies_json,
                status, actual_hours, notes)

-- Notifications
notifications (id, user_id, type, title, body, link_url,
              is_read, created_at)
```

23. API SPECIFICATION

23.1 Base URL & Auth

Base URL: /api/v1
Auth: Bearer JWT in Authorization header

23.2 Key Endpoints

```
AUTH
POST    /auth/login
POST    /auth/google
POST    /auth/refresh
POST    /auth/logout
POST    /auth/forgot-password
POST    /auth/reset-password

ORGANIZATIONS
GET      /orgs/current
PUT      /orgs/current
POST     /orgs/members/invite
GET      /orgs/members
PUT      /orgs/members/{user_id}/role
DELETE   /orgs/members/{user_id}

PROJECTS
GET      /projects
POST     /projects
GET      /projects/{id}
PUT      /projects/{id}
DELETE   /projects/{id}
GET      /projects/{id}/members
POST     /projects/{id}/members

WORKBOOKS
GET      /projects/{pid}/workbooks
POST     /projects/{pid}/workbooks
GET      /workbooks/{id}
PUT      /workbooks/{id}
DELETE   /workbooks/{id}
PUT      /workbooks/{id}/status

PDF PARSING
POST     /workbooks/{id}/upload      -- Upload PDF
POST     /workbooks/{id}/parse      -- Trigger AI parse
GET      /workbooks/{id}/parse/status -- Polling endpoint
GET      /workbooks/{id}/parse/preview -- Structured preview before confirm
POST     /workbooks/{id}/parse/confirm -- Confirm and create sections

SECTIONS
GET      /workbooks/{id}/sections
GET      /sections/{id}
PUT      /sections/{id}
PUT      /sections/{id}/status
POST     /sections/{id}/specs      -- Add spec item
PUT      /sections/{id}/specs/{sid} -- Update spec item
DELETE   /sections/{id}/specs/{sid}
POST     /sections/{id}/regenerate-summary -- Trigger AI re-summary
POST     /sections/{id}/parallax    -- Generate Anime.js visualization

COMMENTS
GET      /sections/{id}/comments
POST     /sections/{id}/comments
```

```
PUT    /comments/{id}
PUT    /comments/{id}/resolve
DELETE /comments/{id}

BOM

GET    /workbooks/{id}/bom
POST   /workbooks/{id}/bom
PUT    /workbooks/{id}/bom/{item_id}
DELETE /workbooks/{id}/bom/{item_id}
POST   /workbooks/{id}/bom/import      -- Import from CSV

QUOTATIONS

GET    /workbooks/{id}/quotations
POST   /workbooks/{id}/quotations
GET    /quotations/{id}
PUT    /quotations/{id}
POST   /quotations/{id}/export        -- Generate document

REVISIONS

GET    /workbooks/{id}/revisions
POST   /workbooks/{id}/revisions/save
GET    /revisions/{id}
POST   /revisions/{id}/activate
POST   /revisions/{id}/lock

INVENTORY

GET    /inventory?scope=org|project&project_id=X
POST   /inventory
PUT    /inventory/{id}
DELETE /inventory/{id}
POST   /inventory/import
GET    /inventory/export

EXPORTS

POST   /workbooks/{id}/export/design-plan
POST   /workbooks/{id}/export/bom
POST   /workbooks/{id}/export/quotation
POST   /workbooks/{id}/export/schedule
POST   /workbooks/{id}/export/spec-sheet
GET    /exports/{export_id}/download

SETTINGS

GET    /settings/user
PUT    /settings/user
GET    /settings/ai
PUT    /settings/ai
GET    /settings/storage
PUT    /settings/storage

NOTIFICATIONS

GET    /notifications
PUT    /notifications/{id}/read
PUT    /notifications/read-all

ADMIN (superadmin only)
```

GET	/admin/organizations
GET	/admin/analytics
PUT	/admin/orgs/{id}/plan
PUT	/admin/feature-flags
GET	/admin/system/health

24. TECH STACK & IMPLEMENTATION GUIDE

24.1 Frontend

Framework:	React 18 + Vite + TypeScript
Styling:	Tailwind CSS 3
Animation:	Framer Motion (transitions/UI) + Anime.js (viz)
State:	Zustand (global) + React Query v5 (server state)
Routing:	React Router v6
WYSIWYG:	TipTap 2 (with StarterKit + custom extensions)
Tables:	TanStack Table v8
PDF Viewer:	react-pdf + PDF.js
Charts:	Recharts (KPIs, progress) + D3 (advanced viz)
Forms:	React Hook Form + Zod validation
Icons:	Lucide React
Date/Time:	date-fns
File Upload:	react-dropzone
Drag & Drop:	@dnd-kit/core
Notifications:	React Hot Toast (transient) + custom drawer
WebSocket:	native WebSocket or Socket.io-client
HTTP Client:	Axios with interceptors
Auth:	@react-oauth/google + custom JWT hooks

24.2 Backend

Framework:	Python 3.11 + FastAPI
Task Queue:	Celery + Redis
Cache:	Redis
PDF Processing:	PyMuPDF (fitz) — primary pdf2image + Pillow — image extraction PyPDF2 — metadata extraction
Image Analysis:	OpenCV
OCR Fallback:	EasyOCR + Tesseract
AI Orchestration:	LangChain (multi-model routing) LlamaIndex (document indexing/retrieval)
Storage:	MinIO (S3-compatible) Google Drive API (google-api-python-client) Firebase Admin SDK
Auth:	python-jose (JWT) + passlib (bcrypt) + google-auth
Email:	FastAPI-Mail + SendGrid/SMTP
WebSocket:	FastAPI native WebSocket
Export:	python-docx (DOCX) + reportlab/weasyprint (PDF) openpyxl (Excel)
DB ORM:	SQLAlchemy 2 + Alembic migrations

24.3 Infrastructure

Database:	PostgreSQL 15
Cache/Queue:	Redis 7
Object Storage:	MinIO (self-hosted) or AWS S3
Reverse Proxy:	Nginx
Containerization:	Docker + Docker Compose
Process Manager:	Supervisor (Celery workers)
Monitoring:	Prometheus + Grafana (optional)
Logging:	Structlog + ELK stack (optional)

24.4 Environment Configuration


```
# .env (application root)

# Database
DATABASE_URL=postgresql://user:pass@localhost:5432/forge

# Redis
REDIS_URL=redis://localhost:6379/0

# Security
SECRET_KEY=<random 64-char hex>
JWT_EXPIRE_MINUTES=15
JWT_REFRESH_DAYS=30

# Google OAuth (optional - app falls back to email/password if not set)
GOOGLE_CLIENT_ID=
GOOGLE_CLIENT_SECRET=

# App-Level AI Keys (used for trial orgs)
OPENAI_API_KEY=
ANTHROPIC_API_KEY=
XAI_API_KEY=
GOOGLE_AI_API_KEY=
PERPLEXITY_API_KEY=
DEEPSEEK_API_KEY=
HUGGINGFACE_TOKEN=

# Storage
MINIO_ENDPOINT=localhost:9000
MINIO_ACCESS_KEY=
MINIO_SECRET_KEY=
MINIO_BUCKET=forge-files

# Firebase (free tier fallback)
FIREBASE_PROJECT_ID=
FIREBASE_CREDENTIALS_JSON=

# Email
SMTP_HOST=
SMTP_PORT=587
SMTP_USER=
SMTP_PASSWORD=

# App Config
SUPERADMIN_EMAIL=admin@yourdomain.com
APP_URL=http://localhost:3000
CORS_ORIGINS=http://localhost:3000

# Feature Flags Override (comma-separated)
FORCE_ENABLE_FEATURES=
FORCE_DISABLE_FEATURES=
```

25. FREE VS PAID FEATURE MATRIX

Feature	Free	Starter	Pro	Enterprise
Organizations	1	1	1	Unlimited
Users	3	10	25	Unlimited
Active Projects	2	10	Unlimited	Unlimited
Workbooks/Project	5	Unlimited	Unlimited	Unlimited
Storage	500MB	5GB	25GB	Custom
AI Calls/Day	10 (HF only)	100 (own key)	Unlimited (own key)	Unlimited
PDF Intelligence	Basic OCR	Full AI parse	Full + Vision	Full + Custom models
Quotation Builder	Basic	Full	Full + Templates	Full + Custom
Inventory	Project only	Org + Project	Full	Full
Revision Control	5 revisions	50	Unlimited	Unlimited
Parallax Views	—	Basic	Full Anime.js	Full + Custom
Export Formats	PDF only	PDF + DOCX + CSV	All formats	All + Custom
Custom Branding	—	Logo only	Full branding	White-label
Google Drive	—	✓	✓	✓
Advanced AI Models	—	Own key	Own key	App key option
Manufacturing Schedule	Basic	Full	Full	Full
Client Portal	—	—	—	✓
API Access	—	—	✓	✓
SSO/SAML	—	—	—	✓
SLA	—	—	99.5%	99.9%
Support	Community	Email	Priority	Dedicated

26. IMPLEMENTATION PHASES

Phase 1 — Foundation (Weeks 1-4)

- Authentication system (email/password + Google OAuth)
- Organization creation + team management
- Project CRUD + basic dashboard
- File upload infrastructure (MinIO/S3)
- Basic PDF display (no parsing yet)
- UI design system implementation (Tailwind + theme)
- Database schema + migrations

Phase 2 — Core Workbook (Weeks 5-8)

- Workbook creation flow
- PDF Intelligence Parser (Stage 1-3)
- Section extraction + display
- Basic spec table (editable)
- Revision control (basic)
- BOM table
- Export: basic PDF

Phase 3 — AI Integration (Weeks 9-12)

- AI Gateway implementation
- Full PDF semantic extraction (Stage 4-6)
- AI model configuration UI
- Section summaries + suggestions
- Weld symbol decoder
- WYSIWYG editor (TipTap)
- Smart Tables (TanStack)

- Review comments system

Phase 4 — Quotation & Inventory (Weeks 13-16)

- Inventory management (org + project)
- Full quotation builder
- Manufacturing schedule
- Labour rate configuration
- Cost breakdown + document export
- Email notifications

Phase 5 — Visualization & Polish (Weeks 17-20)

- Anime.js Parallax visualization engine
- AI-generated visualization from section data
- Advanced revision control (diff viewer)
- Full export suite (DOCX, Excel, PDF)
- Company branding in exports
- Real-time collaboration (WebSocket)
- Performance optimization

Phase 6 — Admin & Monetization (Weeks 21-24)

- Admin panel
- Subscription management
- Feature flags system
- Google Drive sync
- Firebase free-tier storage
- End-to-end testing
- Security audit
- Production deployment

APPENDIX A: MECHANICAL MANUFACTURING PROCESS REFERENCE

The following manufacturing stages should be reflected in the workbook workflow and scheduling:

1. Engineering Review & Planning

- Drawing interpretation, tolerance analysis, material selection confirmation
- Work breakdown structure (WBS) creation
- Jig/fixture requirements assessment

2. Material Procurement

- Material requisition from BOM
- Mill certificate requirements (traceability)
- Incoming inspection

3. Cutting & Preparation

- Plasma/laser/oxy-fuel cutting
- Sawing, shearing
- Edge preparation for welding (beveling, grinding)
- Drilling, punching

4. Fit-Up & Assembly (Box/Tack Stage)

- Dimensional check against drawing
- Jig/fixture setup
- Tack welding per sequence
- Distortion control planning

5. Welding

- Weld procedure specification (WPS) compliance
- Welder qualification records (WQR)
- Preheat requirements (material/thickness dependent)
- Weld sequence for distortion control
- Inter-pass temperature control
- Visual inspection during welding

6. Post-Weld Treatment

- Stress relief (PWHT) if required
- Straightening/dressing
- Grinding, chipping, wire brushing

7. Non-Destructive Testing (NDT)

- Visual inspection (VT)
- Radiographic testing (RT)
- Ultrasonic testing (UT)
- Magnetic particle testing (MT)
- Dye penetrant testing (PT)
- Per drawing/code requirements

8. Dimensional Inspection

- Verification of all critical dimensions
- Flatness, squareness, hole locations
- As-built drawing/sketch

9. Surface Treatment

- Blast cleaning (Sa 2.5 / SSPC-SP6 etc.)
- Priming
- Intermediate coats
- Topcoat
- DFT measurement + records

10. Final Inspection & Documentation - Final dimensional check - Paint DFT records - NDT certificates - Material traceability records - Welder continuity records - Shipping mark / tag

11. Packing & Shipping - Shipping list - Protection (masking, crating) - Rigging point verification

APPENDIX B: DRAWING SECTION TYPES REFERENCE

Common section types FORGE will detect and categorize:

- **Plan View / Top View:** Looking down
- **Elevation View:** Front/side view

- **Section (A-A, B-B, etc.):** Cut view through assembly
- **Detail (Scale X:1):** Enlarged view of specific area
- **Isometric View:** 3D representation
- **Title Block:** Project metadata, revision table, drawing numbers
- **General Notes:** Global notes applying to entire drawing
- **Bill of Materials / Parts List:** Tabulated component list
- **Weld Map:** Weld symbol key/legend
- **Revision Block:** History of drawing changes
- **Match Lines:** Where drawing continues on another sheet

End of FORGE Product Specification v1.0 Document prepared for AI agent implementation All features subject to technical review before implementation